

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	05 July 2026
Team ID	Solo Project – Individual Capstone
Project Name	Library Management System (ServiceNow)
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Requirement Analysis & Planning	USN-1	As a developer, I define the business objectives for the Library Management System.	2	High	Siv (Solo)
Sprint-1		USN-2	As a developer, I define the functional scope (book management, borrow requests, workflow automation, RBAC, reporting).	2	High	Siv (Solo)
Sprint-1		USN-3	As a developer, I map stakeholders (students, librarians, admins, management, IT support).	2	Medium	Siv (Solo)
Sprint-1		USN-4	As a developer, I lay out the 10-phase execution roadmap for the build.	5	High	Siv (Solo)
Sprint-2	Backend Development & Configurations	USN-5	As an admin, I create the Student and Librarian roles.	1	High	Siv (Solo)
Sprint-2		USN-6	As an admin, I create the Book (u_book) and Borrow Request (u_borrow_request) tables.	1	High	Siv (Solo)
Sprint-2		USN-7	As an admin, I add Title, Author, ISBN, and Status fields to the Book table.	2	High	Siv (Solo)
Sprint-2		USN-8	As an admin, I add Requested By, Book, Request Date, and Status fields to the Borrow Request table.	2	High	Siv (Solo)
Sprint-2		USN-9	As an admin, I configure a Reference Qualifier so only Available books can be requested.	2	High	Siv (Solo)
Sprint-2	Development	USN-10	As a developer, I build the Borrow Request Approval Flow in Flow Designer (approve → issue book → notify).	3	High	Siv (Solo)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-2		USN-11	As a developer, I apply a UI Policy to make Return Date mandatory only when a book is Issued.	2	Medium	Siv (Solo)
Sprint-3	Data Migration, Testing & Security	USN-12	As an admin, I configure ACLs on the Book table (read for all, write/delete for Librarian).	2	High	Siv (Solo)
Sprint-3		USN-13	As an admin, I configure ACLs on the Borrow Request table (create/read for Student, write/delete for Librarian).	2	High	Siv (Solo)
Sprint-3		USN-14	As a librarian, I view a "Most Borrowed Books" report grouped by Book.	2	Medium	Siv (Solo)
Sprint-4	Deployment, Documentation & Final Presentation	USN-15	As a developer, I troubleshoot flow execution, ACLs, UI Policy, and reference qualifier behaviour end-to-end.	2	High	Siv (Solo)
Sprint-4		USN-16	As a developer, I track adherence to the planned milestones and timelines.	2	Medium	Siv (Solo)
Sprint-4		USN-17	As a developer, I document the innovation aspects (automation, optimized request handling, scalability).	1	Low	Siv (Solo)
Sprint-4		USN-18	As a developer, I document the technical blueprint (variables, approval use cases, fulfillment tasks).	1	Medium	Siv (Solo)
Sprint-4		USN-19	As a developer, I write the setup manual for recreating the solution on a new PDI.	1	Medium	Siv (Solo)
Sprint-4		USN-20	As a developer, I plan the project demo video walkthrough.	2	Medium	Siv (Solo)
Sprint-4		USN-21	As a developer, I capture visual evidence of forms, approvals, and flow logs.	1	Medium	Siv (Solo)
Sprint-4		USN-22	As a developer, I write clear, sequential explanations (What → Why → How → Outcome) for the final report.	2	Medium	Siv (Solo)
Sprint-4	Project Conclusion	USN-23	As a developer, I summarize the completed solution and outline future scalability.	2	Medium	Siv (Solo)

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	11	3 Days	20 June 2026	22 June 2026	11	22 June 2026
Sprint-2	13	3 Days	23 June 2026	25 June 2026	13	25 June 2026
Sprint-3	6	2 Days	26 June 2026	27 June 2026	6	27 June 2026
Sprint-4	14	7 Days	28 June 2026	04 July 2026	10	In Progress

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = \frac{20}{10} = 2$$

Our Project's Actual Velocity: Total Story Points = 44 across 4 sprints → Velocity = $44 / 4 = 11$ Story Points per Sprint.

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

<https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>

<https://www.atlassian.com/agile/tutorials/burndown-charts>

Reference:

<https://www.atlassian.com/agile/project-management>

<https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software>

<https://www.atlassian.com/agile/tutorials/epics>

<https://www.atlassian.com/agile/tutorials/sprints>

<https://www.atlassian.com/agile/project-management/estimation>

<https://www.atlassian.com/agile/tutorials/burndown-charts>